

Experiment # 02 Implementation of Class & Object in C++

Objectives: To familiarize students working with classes, its objects, data attributes and member functions.

Tools: Dev C++ IDE

Procedure: Write C++ Program to perform the following tasks

- Declare a class with private integer data attribute
- Two member functions (setData, getData) as public
- Declare object of class, and call member function setData to set data
- Call member function getData to show data to the user

The main purpose of C++ programming is to add object orientation to the C programming language and classes are the central feature of C++ that supports object-oriented programming and are often called user-defined types.

Class – is a user defined data type, blue print or template consists of data attributes (variables) and member functions. By using blue print or template means as class can have considered as a house of which functions will be the doors and variables will be items inside house. The functions usually will be the only way to modify the variables in this structure, and they are usually the only way even to access the variables in this structure. The key idea is that the outside world doesn't need to know exactly what data is stored inside the class--it just needs to know which functions it can use to access that data. This allows the implementation to change more easily because nobody should have to rely on it except the class itself.

The syntax for classes is simple. First, you put the keyword '**class**' then the name of the class. Then you put an open bracket. Before putting down the different variables, it is necessary to put the degree of restriction on the variable. There are three levels of restriction. The first is public, the second protected, and the third private. For now, we need to know is that the public restriction allows any part of the program, including parts outside the class, to access the functions and variables specified as public. The private restriction prevents functions outside the class to access the variable. The syntax for declaring these access restrictions using the restriction keyword (public, private) and then a colon. Finally, put the different variables and then you put a closing bracket and semicolon. Classes can also be defined by using “separating interface from implementation” method in which we only declare member functions of a class inside the class and definition of a class outside the class using scope resolution operator (::). In order to define function outside the class, the following syntax is used:

```
return_type class_name :: function_name() {  
    Statements;  
}
```

The idea of declaring something private than public is that some parts of the class are intended to be internal to the class - only for the purpose of implementing features. On the other hand, some parts of the class are supposed to be available to anyone using the class - these are the public class functions. The real world example can be taken as of a microwave: the public parts of the class correspond to the parts of the microwave that you can use on an everyday basis--the keypad, the start button, and so forth. On the other hand, some parts of the microwave are not easily accessible, but they are no less important--it would be hard to get at the microwave generator. These would correspond to the private parts of the class--the things that are necessary for the class to function, but that nobody who uses the class should need to know about.

Object – an instance of a class through which we access member functions of a class. The public data members of objects of a class can be accessed using the direct member access operator (.) such as `object.member_function()`. Private members cannot be accessed directly using direct member access operator (.).

Step 1: Create New C++ Project and save it as Lab_two as shown in the following figures from 2 (a) to 2(d).

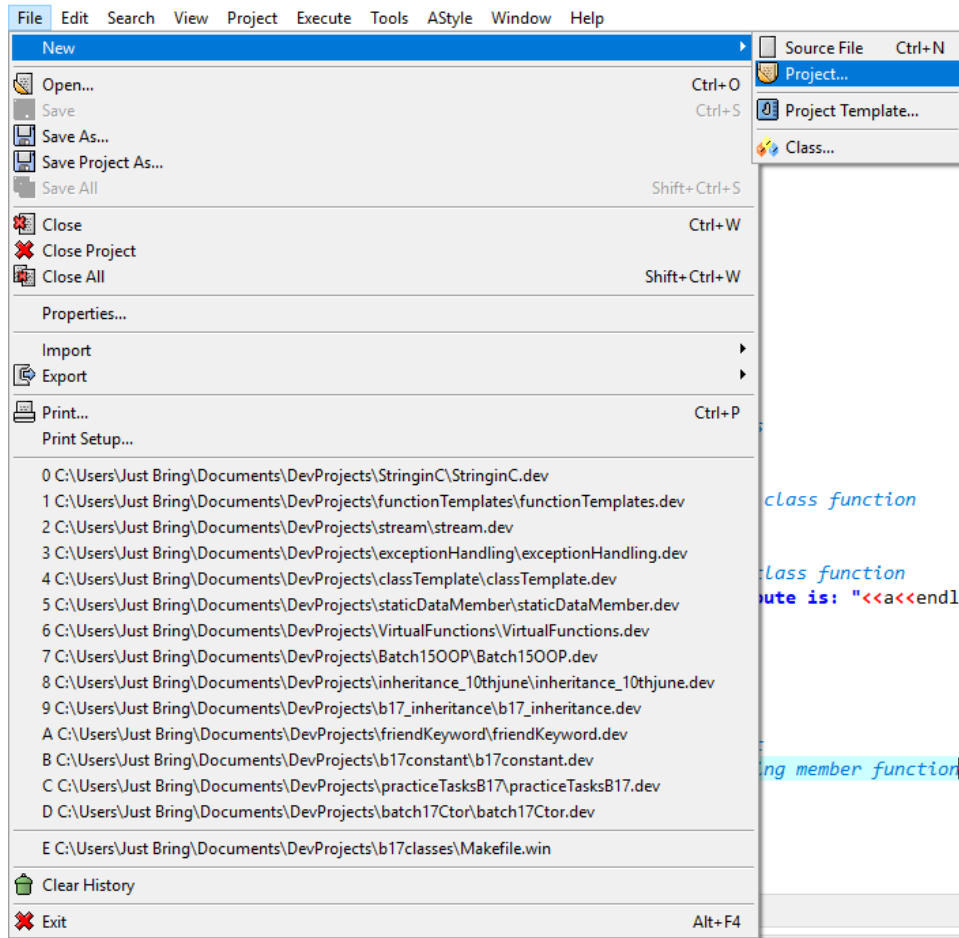


Figure 2 (a): Creating of New Project

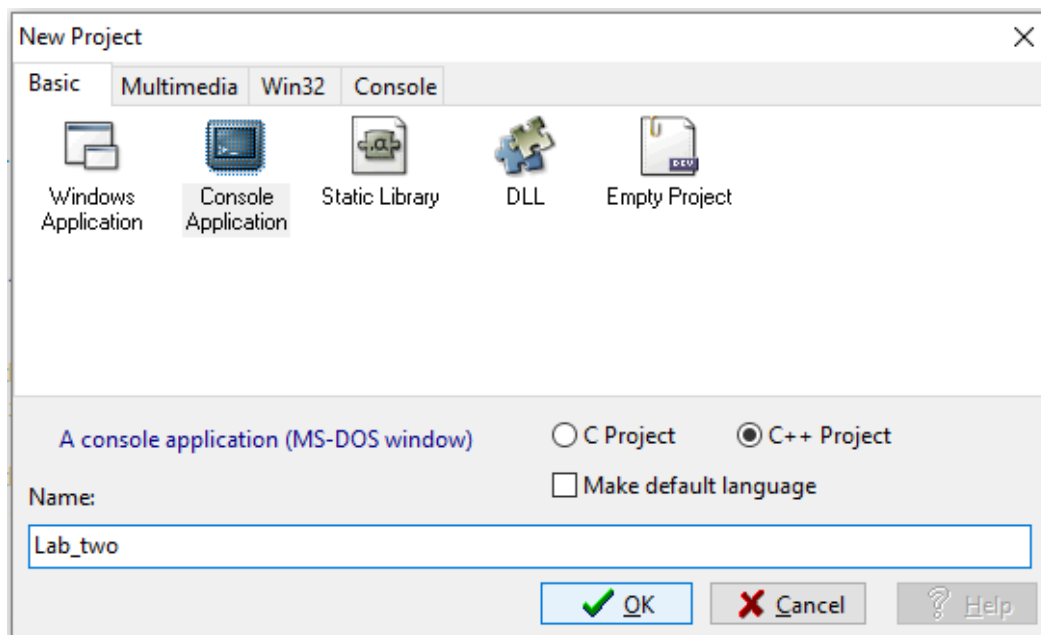


Figure 2 (b): Project Name

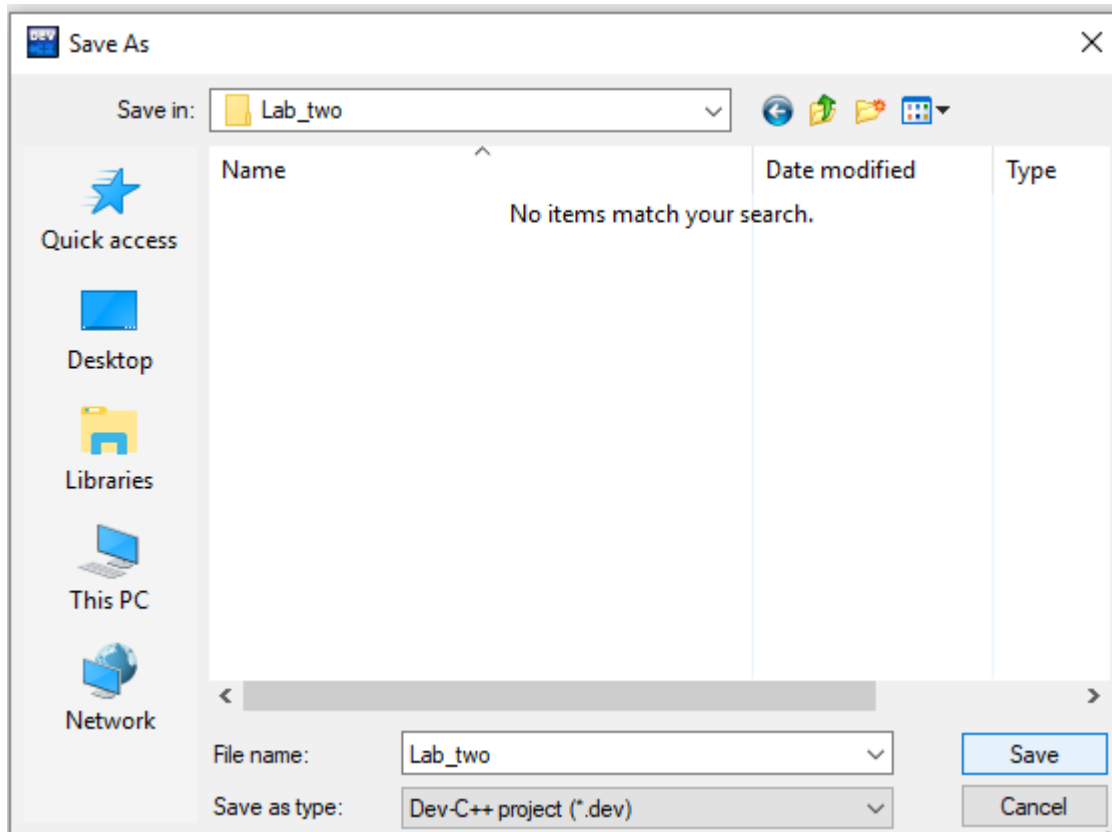


Figure 2 (c): Project Folder

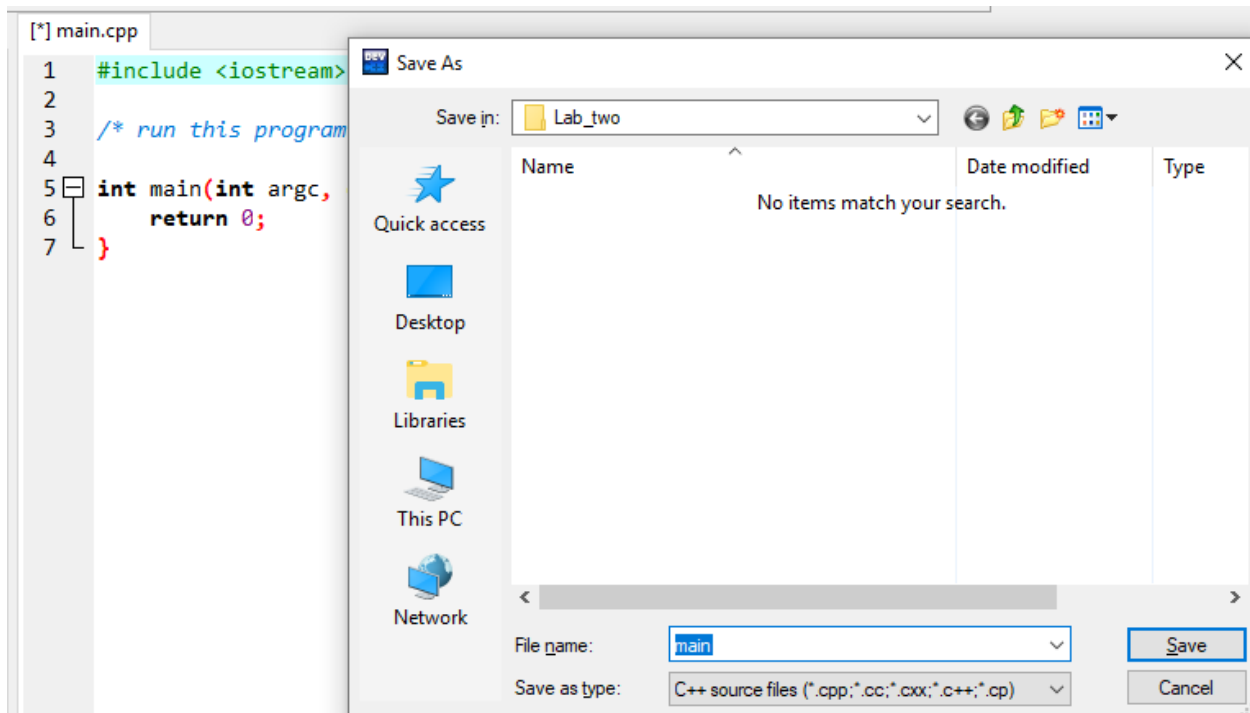
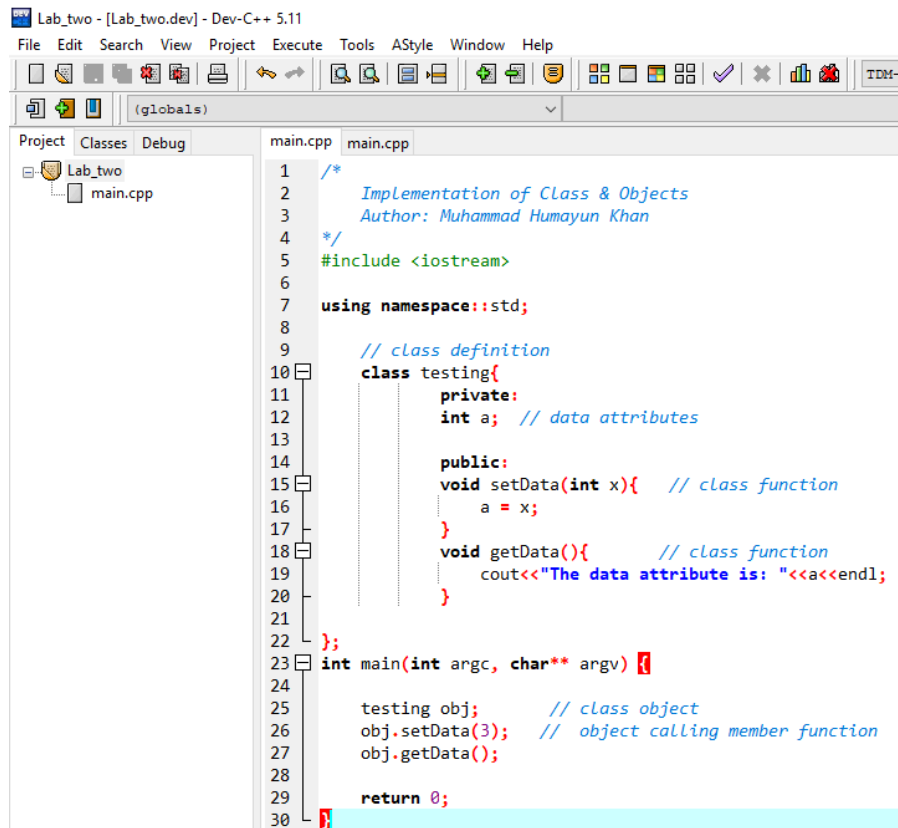


Figure 2 (d): Project Main File

Step 2: Write the following code in main.cpp as shown in figure 2 (e) below and execute to see the desired output.



```
1  /*
2   Implementation of Class & Objects
3   Author: Muhammad Humayun Khan
4  */
5  #include <iostream>
6
7  using namespace::std;
8
9  // class definition
10 class testing{
11     private:
12         int a; // data attributes
13
14     public:
15         void setData(int x){ // class function
16             a = x;
17         }
18         void getData(){ // class function
19             cout<<"The data attribute is: "<<a<<endl;
20         }
21
22 };
23 int main(int argc, char** argv) {
24
25     testing obj; // class object
26     obj.setData(3); // object calling member function
27     obj.getData();
28
29     return 0;
30 }
```

Figure 2 (e): main.cpp source code

Step 3: Next method to implement the above code is called as “separate interface from implementation”. In this method, there are three files such as main.cpp file where the header file of the class is accessed and executed. The other two files are called as header file (.h) and header file implementation file (.cpp). All the declarations of the class attributes and data member functions are in declared in the header file while the implementation of the header file is write in the .cpp file of the header file. The whole process along with the execution is shown below from figure 2 (f) to 2(m).

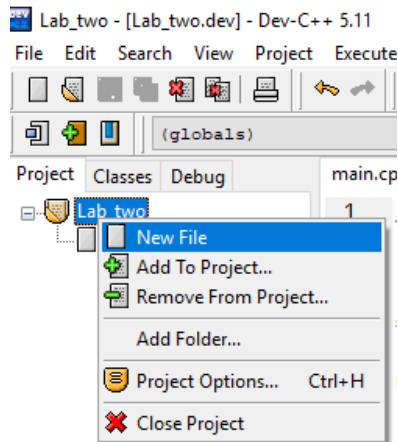


Figure 2 (f): Creating header file

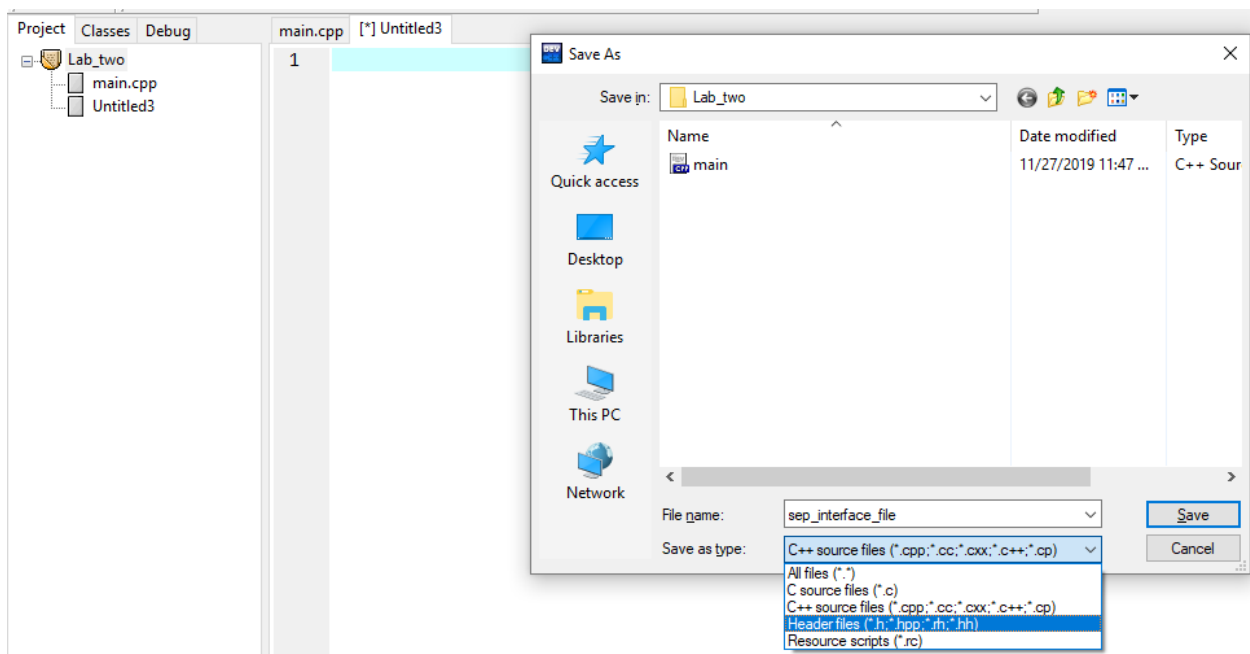


Figure 2 (g): header file name

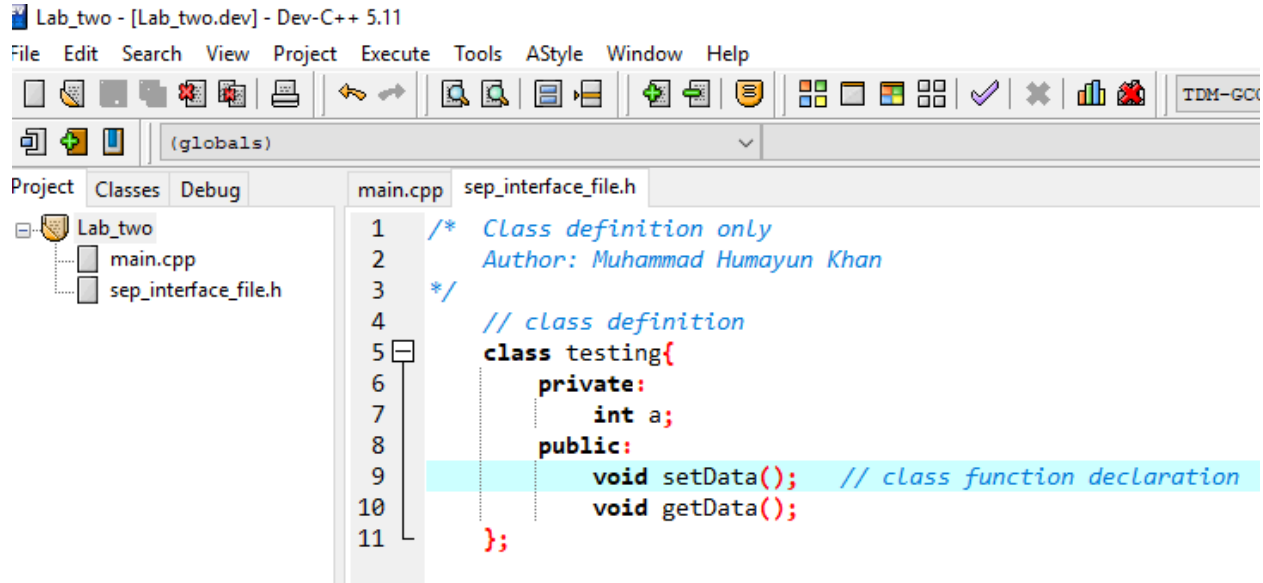


Figure 2 (h): header file code

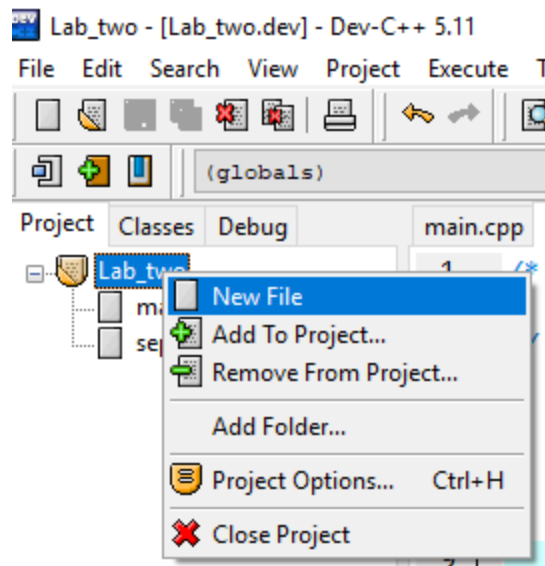


Figure 2 (i): Creating header file .cpp file

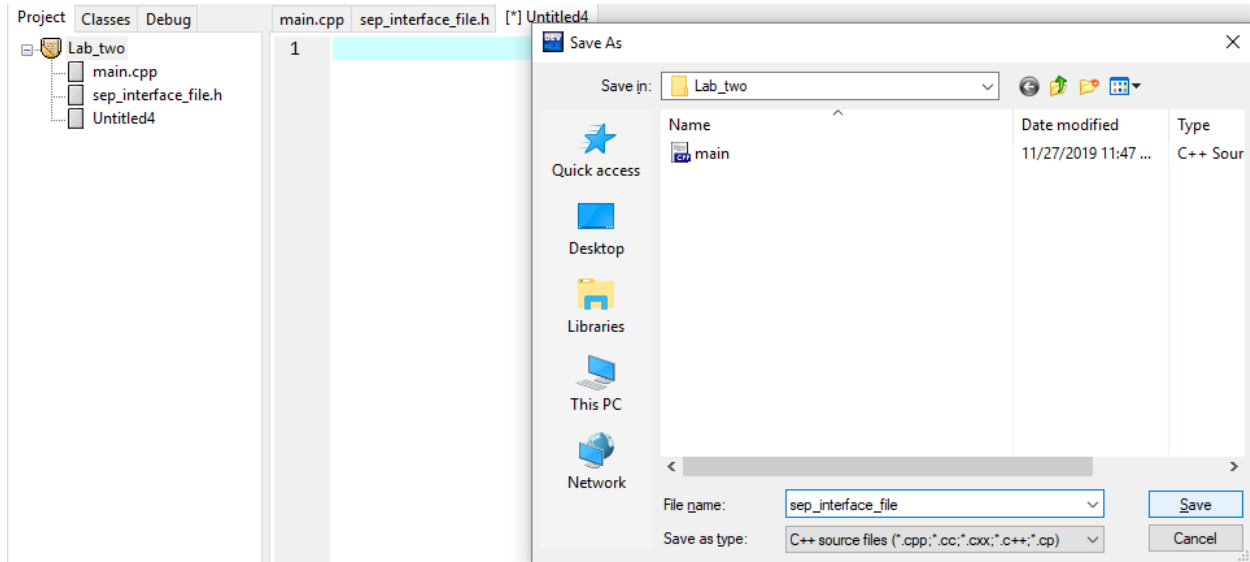


Figure 2 (j): header file .cpp name

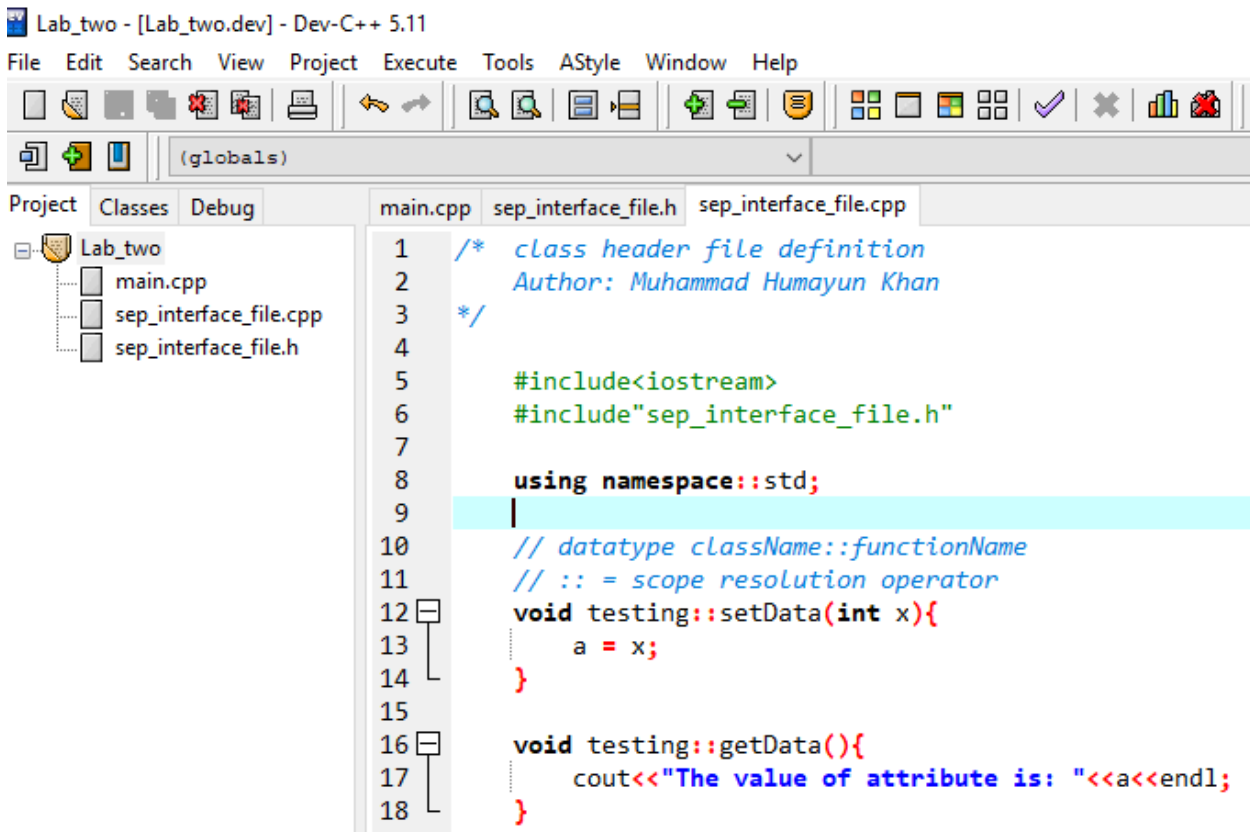


Figure 2 (k): header file .cpp source code

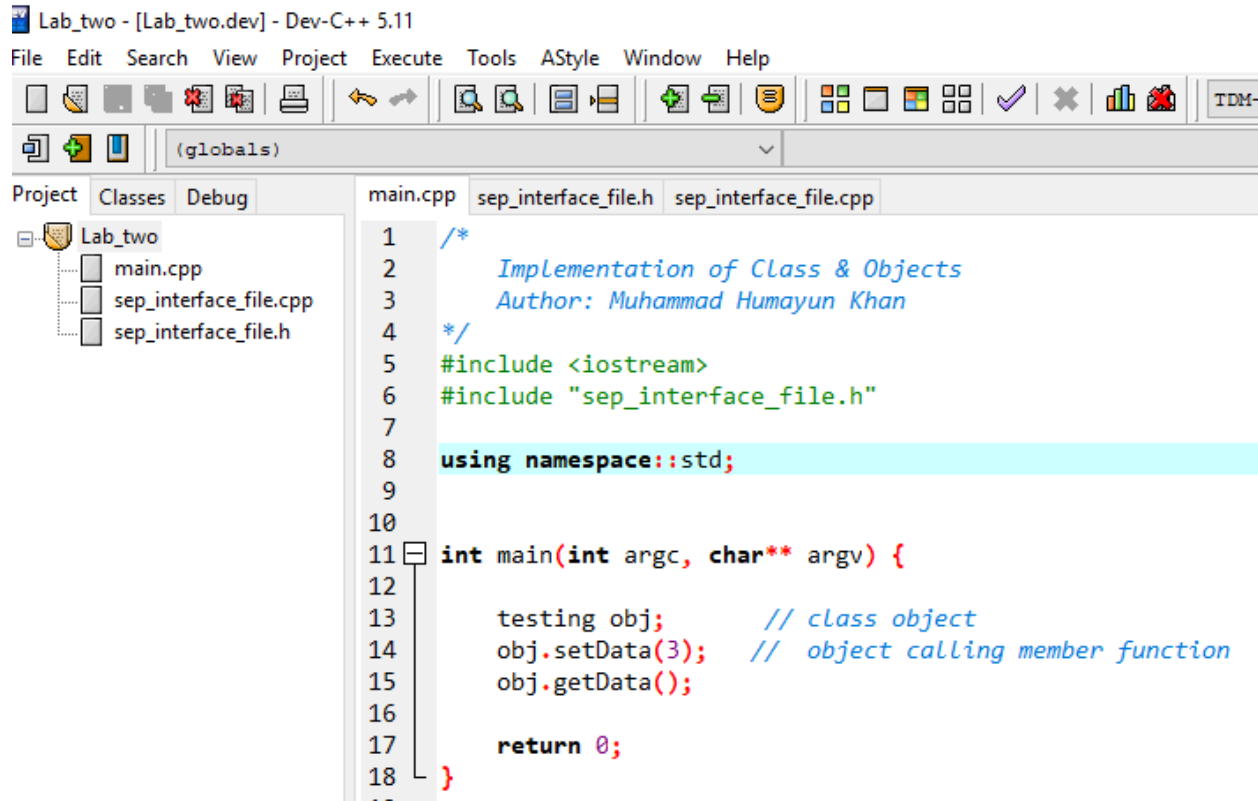


Figure 2 (l): main.cpp source code

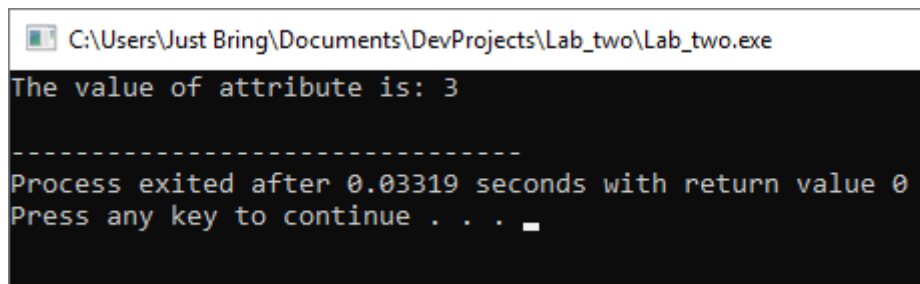


Figure 2(m): Output

Lab Task(s): Write C++ programs to perform following tasks

1. Create a class as **testing** having private data attribute **int a,b** and public members such **void setData(int x) & void getData()**. The setData(int) function should initialize the private data attribute **int a** and should ask user for the value to initialize the data attribute **int b**. The getData() method should display the private data attributes value. Create instance of the class named as **obj** and access the setData(int) and getData() functions.
2. Implement the concept of class in C++ using **separate interface from implementation** method.

3. Receive input from user as *first number, operator and second number* and based on operator, calculate sum, difference, multiplication or division. Ask user for another operation and exit if user enters character N. Solve according to following parts from a to g.
 - a) Declare a class named as testing with following variables
 - Two variables x & y of type float
 - One variable op of char type
 - One variable res of double type
 - b) Declare the following member functions as public
 - Member function input
 - Member function eval
 - Member functions as sum, sub, mul, div and disp
 - c) Declare t as an object of class testing
 - d) Declare a variable ch of type char
 - e) Call input member function of class testing using object
 - f) Call eval member function of class testing using object
 - g) Ask user for another operation
4. Imagine a tollbooth with a class called toll_Booth. The two data items are of type “unsigned int” to hold the total number of cars, and a type double to hold the total amount of money collected. A member function called payingCar () increments the car total and adds 0.50 to the cash total. Another function called nopayCar(), increments the car total but adds nothing to the cash total. Finally a member function called displays the two totals. Include a program to test this class. This program should allow that user to push one key to count a paying car, and another to count a nonpaying car. Pressing ESC (ASCII value = 27) key should cause the program to print out the total cars and total cash and then exit.
5. Create a class taskOne having integer data attributes id, age, salary and a character array as name[25]. The class should have getData() and displayData() member functions. The getData() member function should receive id, age, salary and name (provide full name such as Muhammad Humayun Khan). The display method should print all the received data from the user on console screen. Create array of 3 objects such as taskOne obj[3]. Call getData() thrice and print respective data by calling displayData() method.